

Generalization of Rational Lambert W Integrals

Given the Lambert W function defined by $W(x)e^{W(x)} = x$, we seek to generalize integrals of the form:

$$\int \frac{g(W(x))}{x(1+W(x))} dx \quad (1)$$

Let $u = W(x)$, which implies $x = ue^u$. Differentiating x with respect to u , we obtain:

$$dx = (e^u + ue^u)du = e^u(1+u)du \quad (2)$$

Since $W(x)e^{W(x)} = x$, it follows that $e^u = \frac{x}{u}$. Substituting this back into our differential equation:

$$dx = \frac{x}{u}(1+u)du \implies \frac{dx}{x(1+u)} = \frac{du}{u} \quad (3)$$

Substituting $u = W(x)$ back into the denominator establishes the rigorous identity:

$$\frac{dx}{x(1+W(x))} = \frac{dW(x)}{W(x)} \quad (4)$$

Theorem

For any continuous function $g(u)$ on $[W(a), W(b)]$ for $a, b \geq 0$:

$$\int_a^b \frac{g(W(x))}{x(1+W(x))} dx = \int_{W(a)}^{W(b)} \frac{g(u)}{u} du \quad (5)$$

Application to the Specific Problem

For the integral:

$$\int_0^e \frac{W(x)}{x(1+W(x))(1+W(x)^2)} dx \quad (6)$$

We define $g(W(x)) = \frac{W(x)}{1+W(x)^2}$, meaning $g(u) = \frac{u}{1+u^2}$. The limits transform as $W(0) = 0$ and $W(e) = 1$.

Applying the theorem:

$$\int_0^1 \left(\frac{u}{1+u^2} \right) \frac{1}{u} du = \int_0^1 \frac{1}{1+u^2} du = \left[\arctan(u) \right]_0^1 = \frac{\pi}{4} \quad (7)$$

Extension to Generalized Higher Powers

We can extend this framework to a generalized product in the denominator up to an arbitrary integer power $n \geq 2$:

$$I_n = \int_0^e \frac{W(x)}{x(1+W(x))(1+W(x)^2)\dots(1+W(x)^n)} dx \quad (8)$$

Using product notation, this integral can be written as:

$$I_n = \int_0^e \frac{W(x)}{x \prod_{k=1}^n (1+W(x)^k)} dx \quad (9)$$

By isolating the $x(1+W(x))$ term for the differential substitution, we can identify our function $g(W(x))$ as:

$$g(W(x)) = \frac{W(x)}{\prod_{k=2}^n (1+W(x)^k)} \implies g(u) = \frac{u}{\prod_{k=2}^n (1+u^k)} \quad (10)$$

Applying the established theorem and utilizing the transformed limits $W(0) = 0$ and $W(e) = 1$, the integral simplifies completely into the rational u -domain:

$$I_n = \int_0^1 \left(\frac{u}{\prod_{k=2}^n (1+u^k)} \right) \frac{1}{u} du = \int_0^1 \frac{1}{\prod_{k=2}^n (1+u^k)} du \quad (11)$$

Explicit Integral Framework up to $n = 10$

While a compact closed-form algebraic expression does not exist for arbitrary n , we can explicitly outline the transformed integrals. For higher bounds of n , exact partial fraction decomposition becomes computationally exhaustive due to roots of unity, and these integrals are typically left in this format for numerical approximation.

$$I_2 = \int_0^1 \frac{1}{1+u^2} du = \frac{\pi}{4} \quad (12)$$

$$I_3 = \int_0^1 \frac{1}{(1+u^2)(1+u^3)} du = \frac{5 \ln(2)}{12} + \frac{\pi}{8} \quad (13)$$

$$I_4 = \int_0^1 \frac{1}{(1+u^2)(1+u^3)(1+u^4)} du \quad (14)$$

$$I_5 = \int_0^1 \frac{1}{(1+u^2)(1+u^3)(1+u^4)(1+u^5)} du \quad (15)$$

$$I_6 = \int_0^1 \frac{1}{\prod_{k=2}^6 (1+u^k)} du \quad (16)$$

$$I_7 = \int_0^1 \frac{1}{\prod_{k=2}^7 (1+u^k)} du \quad (17)$$

$$I_8 = \int_0^1 \frac{1}{\prod_{k=2}^8 (1+u^k)} du \quad (18)$$

$$I_9 = \int_0^1 \frac{1}{\prod_{k=2}^9 (1+u^k)} du \quad (19)$$

$$I_{10} = \int_0^1 \frac{1}{\prod_{k=2}^{10} (1+u^k)} du \quad (20)$$